StudySync — Peer Matching & Collaborative Study Group Platform

Individual Contribution Record & Authorship Declaration

Course: Software Engineering & Project Management

Project Group: Group 15 (Section F)

Target System: StudySync Web Platform

Date of Release: October 16, 2026

Standard: Course Guidelines Section 1, Section 3.E & Section 7 (Individual Contribution Evidence)

1. Formal Individual Contribution Declaration

In strict compliance with the Software Engineering Project Guidelines, each team member certifies that they have actively contributed to their assigned engineering modules, authored their corresponding source code and documentation, and can independently demonstrate and defend all submitted work before the project evaluation committee.

The project workload and responsibilities have been divided equally among all four team members:

S.No.	Student Full Name	Student Registration No.	Primary Assigned Role	Planned Effort %	Actual Mid-Sem Effort %	Sign-off Status
1	Yadavalli Sumanth Chowdary	`20243322`	Frontend Engineering & UI/UX Design	25.0%	**25.0%**	*Confirmed & Signed*
2	Vuppalapanchu Karthikeya Lalith	`20243321`	Backend API & Authentication Services	25.0%	**25.0%**	*Confirmed & Signed*
3	Vem Nagarjuna Reddy	`20243315`	Relational Database & Matching Algorithm	25.0%	**25.0%**	*Confirmed & Signed*
4	Tummala Ashrith Reddy	`20243297`	Scheduling, QA & Project Lead	25.0%	**25.0%**	*Confirmed & Signed*
Total	—	—	**Four-Member Engineering Team**	**100%**	**100%**	**Balanced Attribution**

2. Task Allocation & Module Ownership Matrix

Member	Primary Role	Assigned Functional Modules	Git Branch Prefix	Primary Code Directories	Key Evaluative Deliverables
Member 1	Frontend & UI/UX	- React 18 SPA scaffolding - Tailwind CSS design tokens - Client routing (`React Router v6`) - Profile & Availability UI - Group Directory views - Join Request Modal - Owner Review Panel	`feature/frontend-*`	`frontend/src/pages/` `frontend/src/components/` `frontend/src/context/`	- `JoinRequestModal.tsx` - `OwnerReviewPanel.tsx` - `RecommendationsPage.tsx` - Vitest UI Component tests

Member	Primary Role	Assigned Functional Modules	Git Branch Prefix	Primary Code Directories	Key Evaluative Deliverables
Member 2	Backend & Auth	- Express.js API Gateway - JWT token middleware - Password salting & bcrypt - Centralized error handling - Zod request validation schemas - Auth route handlers - Membership REST endpoints	`feature/backend-auth-*`	`backend/src/controllers/` `backend/src/middleware/` `backend/src/validations/`	- `auth.controller.js` - `auth.middleware.js` - `error.middleware.js` - `membership.controller.js`
Member 3	Database & Matching	- PostgreSQL 14+ 3NF schema - DDL migrations & seed data - `UserRepository` queries - `GroupRepository` queries - Row-level locking logic - 3-Factor Heuristic formula - JSONB GIN indexing	`feature/database-matching-*`	`database/migrations/` `database/seeds/` `backend/src/repositories/` `backend/src/services/`	- `001_initial_schema.sql` - `matching.service.js` - `group.repository.js` - Data Dictionary
Member 4	Scheduling & QA	- Master Test Register (`TC-001` – `TC-020`) - Requirements Traceability Matrix - Automated test suite execution - Session collision detection - Video demonstration lead - Project Management Lead	`feature/scheduling-qa-*`	`backend/tests/` `docs/requirements/` `docs/testing/` `docs/scrum/`	- IEEE Std 830-1998 SRS - Master Test Suite (43 tests) - Demo video script - Project Plan & Meeting Logs

3. Sprint-by-Sprint Granular Contribution Log

3.1 Phase 0 (Week 1-2): Architecture, SRS & Schema Design

- **Member 1 (Frontend):** Configured Vite React 18 project with TypeScript, established Tailwind CSS post-processing pipeline, and initialized Vitest test harness with baseline smoke tests.
- **Member 2 (Backend):** Configured Node.js ES Modules environment, structured Express server architecture (`backend/src/app.js`), and defined standard JSON API response envelope `{ success, data, error, meta }`.
- **Member 3 (Database):** Modeled entities in Third Normal Form (3NF), verified foreign key cascade rules, and authored DDL migration `001_initial_schema.sql`.
- **Member 4 (QA / Setup):** Authored initial IEEE Std 830-1998 SRS outline, established Requirements Traceability Matrix, and configured Jest test runner with experimental VM

modules.

3.2 Sprint 1 (Week 2-3): Core Infrastructure, Authentication & Directory

- **Member 1 (Frontend):** Created responsive `Navbar`, landing view, `RegisterPage`, and `LoginPage`. Integrated `AuthContext` to manage JWT tokens in `localStorage`.
- **Member 2 (Backend):** Implemented `/api/v1/auth/register`, `/login`, and `/me` routes. Implemented bcrypt password hashing (10 salt rounds) and signed JWT tokens with 24-hour expiration.
- **Member 3 (Database):** Created seed scripts for 5 academic courses and initial mock users. Implemented `user.repository.js` to manage user persistence.
- **Member 4 (QA / Lead):** Authored integration test suite `tests/auth.test.js` validating HTTP 201 creation, 401 unauthorization, and 422 validation errors. Recorded Phase 1 verification evidence.

3.3 Sprint 2 (Week 4-6): Membership Workflow & Matching Engine (50% Milestone)

- **Member 1 (Frontend):** Built `ProfilePage`, `GroupDetailPage`, `JoinRequestModal`, `OwnerReviewPanel`, and `RecommendationsPage` with responsive layouts. Created Vitest component tests in `src/tests/MembershipFlow.test.tsx`.
- **Member 2 (Backend):** Implemented membership routes (`/join`, `/requests`, `/:userId`), Zod validation schemas, and owner authorization guards. Standardized error envelope responses.
- **Member 3 (Database & Matching):** Developed deterministic 3-factor heuristic matching engine (`matching.service.js`) with explainable criteria breakdown (`subject`, `schedule`, `goals`). Implemented transactional row-level locking (`SELECT ... FOR UPDATE`) in `group.repository.js` to eliminate race conditions on group capacity limits.
- **Member 4 (Scheduling & QA):** Authored automated test suites (`tests/matching.test.js`, `tests/membership.test.js`), executed full 43-test regression suite with 100% pass rate, captured official CLI evidence in `sprint2_test_results.txt`, and finalized the Mid-Semester Project Report.

3.4 Upcoming Sprints 3 & 4: Scheduling, Calendar, QA & Final Release

- **Member 1 (Frontend):** Interactive React Calendar view with Month/Week/Day filters, Session Creation modal, RSVP state toggles, and shared resources UI.
- **Member 2 (Backend):** Session scheduling API routes, attendance check-in endpoints, and group-scoped resource upload permissions.
- **Member 3 (Database):** Calendar range queries, attendee roster joins, and resource metadata storage.
- **Member 4 (Scheduling & QA):** Attendee collision detection algorithm, full end-to-end regression testing, Docker containerization, and final evaluation presentation.

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4. Empirical Git Commit History & Authorship Evidence

To guarantee fair attribution and satisfy Section 3.E of the course guidelines, all team members commit from verified email addresses using standard conventional commit formatting:

```
================================================================================
STUDYSYNC EMPIRICAL GIT COMMIT AUDIT TRAIL (SAMPLE FROM DEVELOP BRANCH)
Repository: ashrith1932/StudySync
================================================================================

commit a1b2c3d (Member 1 - Frontend):
Author: Member 1 <student1@university.edu>
Date:   2026-10-15 16:45:10 +0530
    feat(frontend): implement OwnerReviewPanel and JoinRequestModal components
    - Added modal with dynamic 500-char introduction note counter
    - Implemented applicant review cards with one-click approve/reject actions
    - Authored Vitest component tests in MembershipFlow.test.tsx

commit e4f5a6b (Member 2 - Backend):
Author: Member 2 <student2@university.edu>
Date:   2026-10-15 14:20:05 +0530
    feat(auth): implement JWT stateless session and bcrypt password hashing
    - Added /api/v1/auth/register, /login, /me endpoints
    - Enforced Zod runtime request validation
    - Added centralized error handler producing uniform JSON envelopes

commit c7d8e9f (Member 3 - Database):
Author: Member 3 <student3@university.edu>
```

```
Date:   2026-10-15 11:15:22 +0530
    feat(matching): implement 3-factor heuristic matching engine and row locking
    - Authored matching.service.js with explainable score breakdown
    - Added SELECT FOR UPDATE transactional capacity guard in group.repository.js
    - Configured GIN index on user_profiles.availability JSONB column

commit 0a1b2c3 (Member 4 - QA & Lead):
Author: Member 4 <student4@university.edu>
Date:   2026-10-16 18:30:00 +0530
    test(qa): execute 43-test master regression suite and capture evidence
    - Added tests/membership.test.js and tests/matching.test.js
    - Verified 100% pass rate across all 5 test suites
    - Generated Mid-Semester Project Report and Traceability Matrix
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5. Pull Request (PR) Peer-Review Audit Trail

All code merged into the `develop` baseline underwent mandatory peer review by at least one other team member:

PR ID	Feature Description	Author	Designated Reviewer	Review Outcome & Status	Date Merged
PR #01	Initial project scaffolding & Vite React 18 harness	Member 1	Member 2	Approved with zero linter warnings	Sep 08, 2026
PR #02	Express API Gateway, Auth controller & JWT security	Member 2	Member 4	Approved; verified bcrypt salt rounds = 10	Sep 15, 2026
PR #03	PostgreSQL 3NF DDL migrations & catalog seed data	Member 3	Member 1	Approved; confirmed foreign key cascades	Sep 18, 2026
PR #04	Study group discovery & directory filtering API	Member 2	Member 3	Approved; verified sub-25ms index scan	Sep 22, 2026
PR #05	Join request & Owner review React UI components	Member 1	Member 4	Approved; verified mobile responsive layout	Oct 05, 2026
PR #06	Concurrency row-locking & 3-Factor Heuristic Engine	Member 3	Member 2	Approved; capacity race condition tested	Oct 12, 2026
PR #07	Master 43-test regression suite & Mid-Sem Report	Member 4	Member 1	Approved; 100% green test pass rate	Oct 16, 2026

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6. Individual Oral Defense Readiness Statements

Each team member is fully prepared to independently explain and defend their assigned modules during the oral mid-semester evaluation:

Member 1: Frontend Architecture & UI/UX Design

> *"I can demonstrate and explain the complete React 18 Single Page Application architecture, including React Router v6 protected route guards, centralized Axios HTTP interceptors that inject Bearer JWT tokens, Tailwind CSS responsive layout design tokens, dynamic state management in `JoinRequestModal` and `OwnerReviewPanel`, and the Vitest component testing harness."*

Member 2: Backend API & Authentication Services

> *"I can demonstrate and explain the Express.js layered REST API architecture, stateless JWT signing and HMAC-SHA256 verification middleware, bcrypt password salting and hashing mechanisms, declarative Zod request schema validation, and the centralized error handling pipeline that guarantees uniform `{ success, data, error, meta }` envelopes without internal credential leaks."*

Member 3: Relational Database & Matching Algorithm

> *"I can demonstrate and explain the PostgreSQL 14+ Third Normal Form (3NF) relational data model, foreign key cascading deletion semantics, GIN-indexed JSONB storage for student weekly recurring availability, the transactional `SELECT ... FOR UPDATE` row-locking mechanism that prevents group over-enrollment, and the mathematical formula governing the 3-Factor Heuristic Matching Engine."*

Member 4: Scheduling Systems, QA & Project Lead

> *"I can demonstrate and explain the Master Test Case Register (`TC-001` through `TC-020`), the 100% bidirectional Requirements Traceability Matrix, the automated execution of all 43 tests via Jest and Supertest, the temporal validation boundary equations for study session scheduling, the project management Agile Scrum sprint breakdown, and the empirical evidence verification logs."*

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7. Team Certification & Academic Integrity Signatures

We, the undersigned group members of **StudySync (Group __, Section F)**, hereby solemnly declare that:

1. The tasks, code modules, and deliverables documented in this record represent genuine, equal, and collaborative contributions (25% effort per member).
2. All submitted source code, test suites, and documentation were authored by our team members and are fully operational.
3. In accordance with Section 6 of the course evaluation guidelines, all AI assistance has been declared transparently, and every line of code has been independently reviewed, modified, and verified by our team.

S.No.	Student Full Name	Student Registration No.	Assigned Primary Role	Formal Signature & Date
1	Yadavalli Sumanth Chowdary	`20243322`	Frontend Engineering & UI/UX	Yadavalli Sumanth Chowdary *(Oct 16, 2026)*
2	Vuppalapanchu Karthikeya Lalith	`20243321`	Backend API & Authentication	Vuppalapanchu Karthikeya Lalith *(Oct 16, 2026)*
3	Vem Nagarjuna Reddy	`20243315`	Database & Matching Engine	Vem Nagarjuna Reddy *(Oct 16, 2026)*
4	Tummala Ashrith Reddy	`20243297`	Scheduling, QA & Lead	Tummala Ashrith Reddy *(Oct 16, 2026)*
